

01

Safe Deep Reinforcement Learning for Multi-Objective Energy Cyber-Physical Systems

TECHNICAL BRIEF | PV-BATTERY ENERGY CYBER-PHYSICAL SYSTEM

A magazine-style synthesis of the methodology, numerical evidence, controller comparison, safety case, computational feasibility and explainable-AI results.

100% Q4 completion with shield	100% Full-year completion	2.806 t Annual CO ₂ complete controller	71.28% Annual self- sufficiency	0.371 ms Mean shield decision time
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EXECUTIVE SYNTHESIS

Main outcome

Forecast-aware PPO improved anticipatory dispatch but remained vulnerable at both low- and high-SOC boundaries. The semi-formal runtime shield restored complete Q4 and full-year operation by checking, clipping or blocking unsafe actions before the Modelica FMU step.

SYSTEM AT A GLANCE

Item	Implementation
Physical plant	Modelica FMU PV-battery microgrid
Co-simulation	Python + FMPy + Gymnasium
Learning controller	Stable-Baselines3 PPO, MlpPolicy
Forecasting	LSTM GHI/load, +1h to +4h
Safety layer	SOC, voltage, ramp and forecast-aware reserve
Evaluation	Three-day, unseen Q4 and full year 2023

EVIDENCE-BASED CLAIMS

- Among complete controllers, the shielded POMDP achieved the strongest Q4 and annual CO₂, SSR, SSC, grid-import, voltage and composite economic outcomes.
- The benefit required substantially more battery activity: annual heat loss was 1,269.21 kWh versus 284.73 kWh for Enhanced RBC.
- The shielded environment required about 1.01 ms per step; the measured shield decision itself required about 0.37 ms.
- Improved local XAI reached mean R² values from 0.950 to 0.974, with action-label agreement from 95.9% to 98.0%.

RESEARCH QUESTIONS RESOLVED BY THE BRIEF

Question	Evidence used
Does forecasting improve anticipation?	Unseen-Q4 LSTM accuracy, POMDP behaviour and forecast-dependence XAI
Does PPO alone ensure safe operation?	Training instability, SOC penalties and Q4/annual boundary failures
Does the shield restore deployability?	100% completion, zero observed violations and measured intervention rates
Is the trustworthy layer computationally feasible?	Mean and p95 timing relative to the 15-minute control interval

takeaway

The thesis contribution is the integration and evaluation of a physical digital twin, forecast-aware PPO, runtime constraint enforcement, trust metrics and quantitatively validated XAI—not a claim that PPO, LSTM or shielding are individually new.

Figure 1: End-to-End Trustworthy ECPS Control Architecture

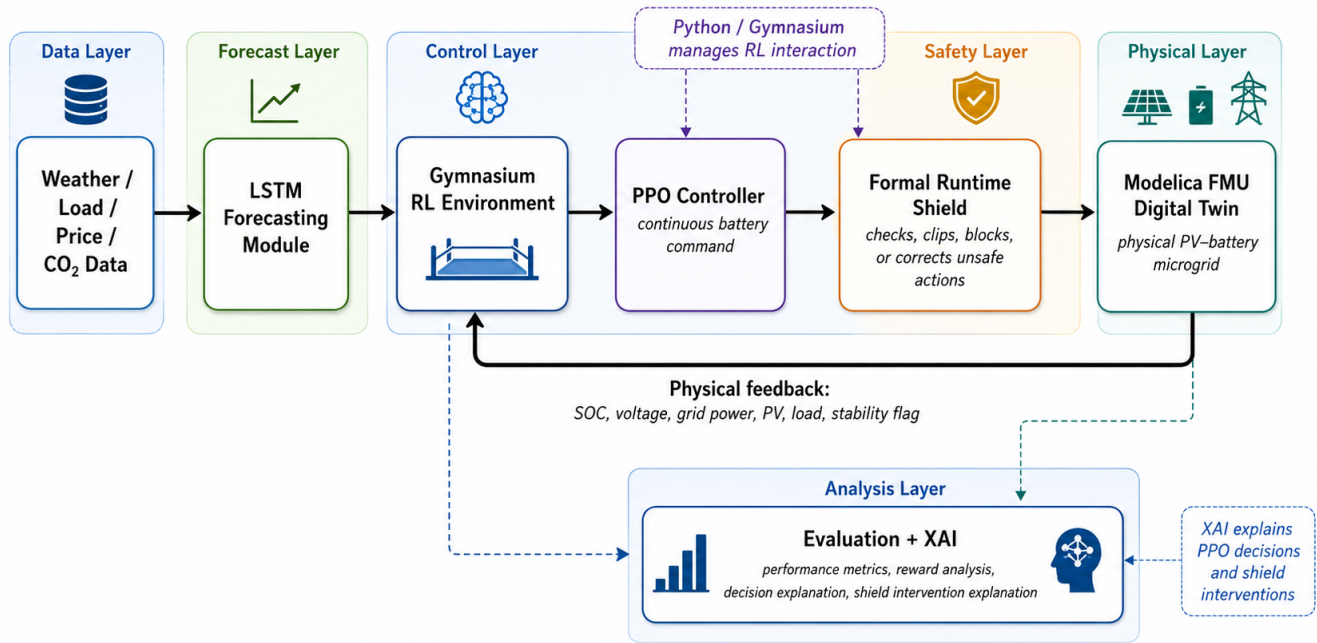


Figure 1. Retained block diagram from the prior brief, updated numerically in the following pages.

LAYER ROLES

Layer	Primary function	Evidence produced
Data	Weather, load, price and CO ₂ inputs	Time-aligned 15-minute operating state
Forecast	LSTM predicts future GHI and load	Q4 MAE/RMSE and persistence comparison
Control	Gymnasium observation and PPO action proposal	Reward, episode length and dispatch behaviour
Safety	Dynamic reserve, safe bounds, clip/block/fallback	Intervention counts, margins and reason codes
Physical	Modelica FMU PV-battery response	SOC, Pgrid, voltage, heat loss, stability_flag
Analysis	Economic, trust, overhead and XAI assessment	Tables, charts, local explanations and diagnostics

Closed-loop principle

The controller never receives a static label. It proposes an action, the digital twin responds physically, and that response determines the next state, reward, diagnostics and explanation.

Trust principle

Optimization and safety are separated. PPO seeks useful dispatch; the shield enforces operational constraints before the FMU executes the command.

CLOSED-LOOP EVIDENCE CHAIN

Stage	Recorded evidence	Decision value
Forecast	GHI/load +1h...+4h and Q4 error metrics	Quantifies anticipatory information
PPO proposal	Continuous battery command and policy checkpoint	Separates learned intent from executed action
Shield decision	Safe bounds, correction, block and reason code	Shows why an action changed or passed
FMU response	SOC, voltage, grid power, heat and stability_flag	Tests physical consistency
Evaluation / XAI	Reward, cost, trust metrics and local fidelity	Connects performance with safety and explanation

Professor-ready takeaway

The architecture supports traceability from input data to forecast, proposed action, shield decision, physical response and human-readable explanation.

03

Digital Twin, FMU Interface and Co-Simulation

PHYSICAL GROUNDING AND FAILURE DIAGNOSTICS

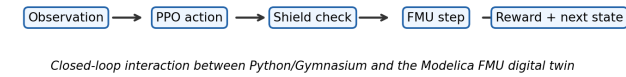
MODELICA DIGITAL-TWIN OUTPUTS

FMU variable	Role
SOC	Battery energy state
P_grid	Grid import/export power
P_pv / P_load	Generation and demand
V_pcc_pu	PCC voltage quality
P_loss_bat	Battery heat/loss proxy
stability_flag	Global numerical/physical validity

PHYSICAL COMPONENTS

- PV array and irradiance-driven generation
- Battery charge/discharge and SOC dynamics
- Grid connection and residential load
- PCC voltage and line behaviour
- Heat-loss proxy for battery stress

PYTHON/GYMNASIUM LOOP



RL element	Definition
Time step	15 minutes
Action	Continuous battery command
MDP observation	12 current-state features
POMDP observation	20 features with GHI/load forecasts
Termination	FMU instability or unsafe condition
Training/testing	Q1–Q3 train; Q4 unseen test

FAILURE DIAGNOSTICS CARRIED FORWARD TO THE PROFESSOR BRIEF

Logged evidence	Interpretive use
SOC, voltage, Pgrid, PV, load	Associates the failure with a physical operating region
Proposed vs executed action	Shows whether unsafe behaviour came from PPO or the shield
stability_flag	Confirms FMU termination but not the unique internal cause
Completion ratio / failure step	Measures operational continuity and deployability
Low- and high-SOC depth	Distinguishes lower-bound from upper-bound failure modes

diagnostic statement

The FMU currently exposes one global stability_flag. The observed failures are strongly associated with SOC-boundary operation while voltage remains within limits, but the exact internal Modelica failure mechanism cannot be uniquely isolated without additional diagnostic flags.

CORE PHYSICAL RELATIONS USED FOR INTERPRETATION

Relation	Interpretation
$P_{grid} = P_{load} - P_{pv} - P_{battery}$	Positive values indicate grid import; negative values indicate export
$SOC_{next} = SOC \pm energy / capacity$	Links a 15-minute battery command to the next SOC
$Voltage\ margin = \min(V - 0.95, 1.05 - V)$	Measures distance to the nearest PCC voltage boundary
$Completion = \frac{executed\ transitions}{expected\ transitions}$	Distinguishes natural horizon completion from early failure

The two failure modes that motivate the shield

Without the shield, the forecast-aware controller crashed in two opposite ways. In Q4 it kept discharging at a very low charge level (SOC 0.149). Over the full year it kept charging at a very high level (SOC 0.850). In both cases voltage was normal, yet the model stopped. Because failures appear at both ends, both the lowest and highest SOC must be reported — a single minimum-SOC number would hide the upper-limit crash.

Failure	When	SOC	Command	Voltage	Result
Lower-bound (Q4)	step 4,374	0.149	still discharging	0.998 pu	FMU stop
Upper-bound (annual)	step 10,915	0.850	still charging	1.001 pu	FMU stop

ay

cal twin is not a visualization layer; it is the physical authority that reveals when a learned battery action leaves the valid operating region.

04

LSTM Forecasting on Unseen Q4

GENERALIZATION EVIDENCE FOR GHI AND LOAD AT +1H TO +4H

LSTM forecasting remains useful as horizon increases

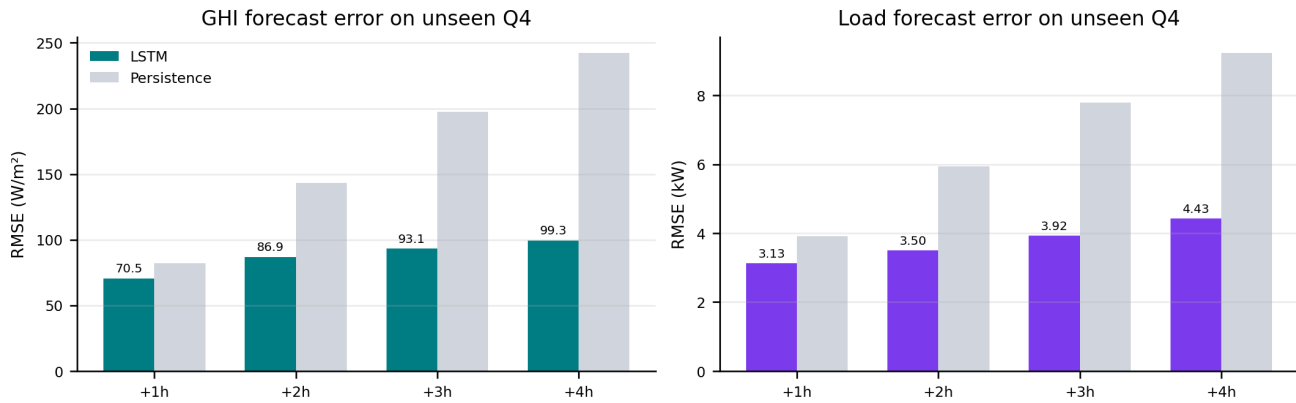


Figure 2. RMSE comparison against persistence on the unseen winter quarter.

Horizon	GHI RMSE	GHI improvement	Load RMSE	Load improvement
+1h	70.491 W/m ²	14.11%	3.129 kW	20.00%
+2h	86.874 W/m ²	39.32%	3.503 kW	41.09%
+3h	93.065 W/m ²	52.90%	3.924 kW	49.68%
+4h	99.297 W/m ²	59.02%	4.427 kW	52.05%

TRAINING BEHAVIOUR

Normalized loss: 0.024916 → 0.003531 in 10 epochs (≈85.8% reduction).

- Largest improvement occurred between epochs 1 and 2.
- Later epochs showed diminishing returns rather than divergence.
- Training loss alone does not prove generalization; the Q4 test is the key evidence.

CONTROL USE BY HORIZON

Forecast	Recommended role
+1h	Immediate battery dispatch
+2h	Short-term scheduling
+3h	Reserve preparation
+4h	Risk warning and dynamic SOC reserve

Interpretation

Forecast error increases with horizon, but persistence deteriorates much faster. The LSTM therefore supplies useful anticipatory information, especially for +2h to +4h planning. Forecasts guide control; they are not hard safety guarantees.

FULL VALIDATION STATISTICS

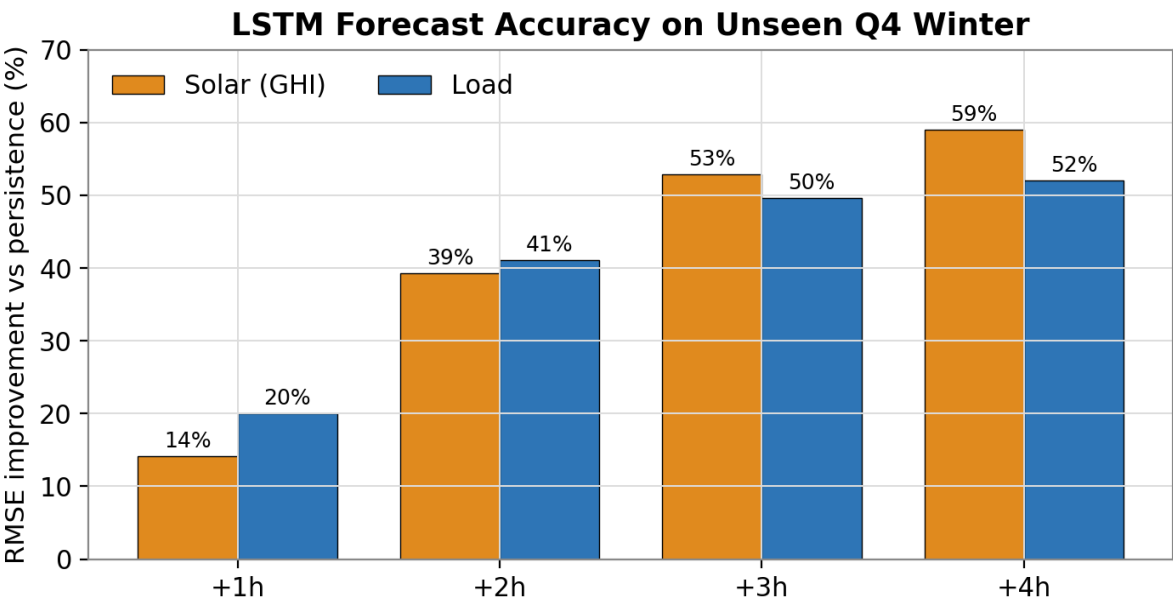
Horizon	GHI MAE	GHI daytime NRMSE	Load MAE	Load NRMSE
+1h	39.067 W/m ²	21.93%	2.443 kW	15.77%
+2h	45.675 W/m ²	27.03%	2.683 kW	17.66%
+3h	48.129 W/m ²	28.96%	2.978 kW	19.79%
+4h	51.476 W/m ²	30.90%	3.320 kW	22.34%

OPERATIONAL INTERPRETATION AND LIMITATIONS

Load is more predictable than GHI in normalized terms. RMSE remains above MAE at every horizon, indicating occasional larger errors during irradiance ramps or demand peaks. The forecasts are therefore useful for scheduling and reserve preparation, but uncertainty should increase with horizon and sharp transitions should remain protected by runtime constraints.

takeaway

The Q4 validation supports adding forecast features to the POMDP observation and to the dynamic reserve logic, while preserving a separate runtime safety layer for forecast error and policy error.



05

Controllers, Observation Design and PPO Configuration

A CONTROLLED METHODOLOGICAL PROGRESSION

THREE LEARNING-BASED CONTROLLERS

Aspect	MDP-PPO	POMDP-LSTM-PPO	Shielded POMDP-LSTM-PPO
Observation	12D current state	20D + LSTM forecasts	20D + LSTM forecasts
Forecast awareness	No	GHI/load +1h...+4h	Forecasts + dynamic reserve
Policy	MlpPolicy	MlpPolicy	MlpPolicy
Runtime verification	Basic environment protection	Basic environment protection	SOC, voltage, ramp, safe-set check
Full blocking	Not observed	Not observed	Yes
Fallback RBC	No	No	Available
Learning penalty	Base reward	Base reward	Correction / block / fallback penalties

CONFIRMED PPO HYPERPARAMETERS

Parameter	Value
Algorithm	Stable-Baselines3 PPO
Policy	MlpPolicy
Timesteps	500,000
Learning rate	0.0003
n_steps	2,048
Batch size	64
Epochs	10
Gamma	0.999
GAE λ	0.95
Clip range	0.2
Seed	42

IMPORTANT METHODOLOGICAL CLARIFICATION

LSTM is external to PPO

The policy is not recurrent. The LSTM predicts GHI and load; those predictions become observation features. PPO still uses a feed-forward MLP to propose the battery action.

ACTION SEMANTICS

Command	Meaning
Negative	Charge battery
Positive	Discharge battery
Range	Approximately -50 to +50 kW
Execution	Original or shield-corrected action reaches FMU

DISCOUNT INTERPRETATION

$\gamma = 0.999$ gives an approximate $1/(1-\gamma) = 1,000$ -step horizon, about 10.4 simulated days at 15-minute resolution. This supports long-term energy consequences but can make value estimation more difficult.

MULTI-OBJECTIVE REWARD PRIORITIES

Reward term	Configured share / role
Economic effect	0.55 — primary operating objective
CO ₂ effect	0.20 — discourages carbon-intensive import
Voltage deviation	0.10 — supports power quality
Battery heat / loss	0.05 — discourages hardware stress
Local energy use	0.10 — rewards PV self-use and autonomy

MODEL-SELECTION RULE

Training duration is not a performance criterion. Each controller is selected from evaluation checkpoints using reward together with completion, minimum SOC, intervention burden and physical metrics. This is why the latest MDP best_eval checkpoint completes Q4 and the full year even though the training process itself was unstable.

takeaway

Because PPO settings are held constant, the MDP-to-POMDP comparison primarily tests forecast information; the shielded controller additionally changes execution and learning through runtime constraint enforcement.

06

Training Convergence and Numerical Output

FORECASTING HELPS; RUNTIME ENFORCEMENT STABILIZES

Forecasts improve average behaviour; the runtime shield changes convergence qualitatively

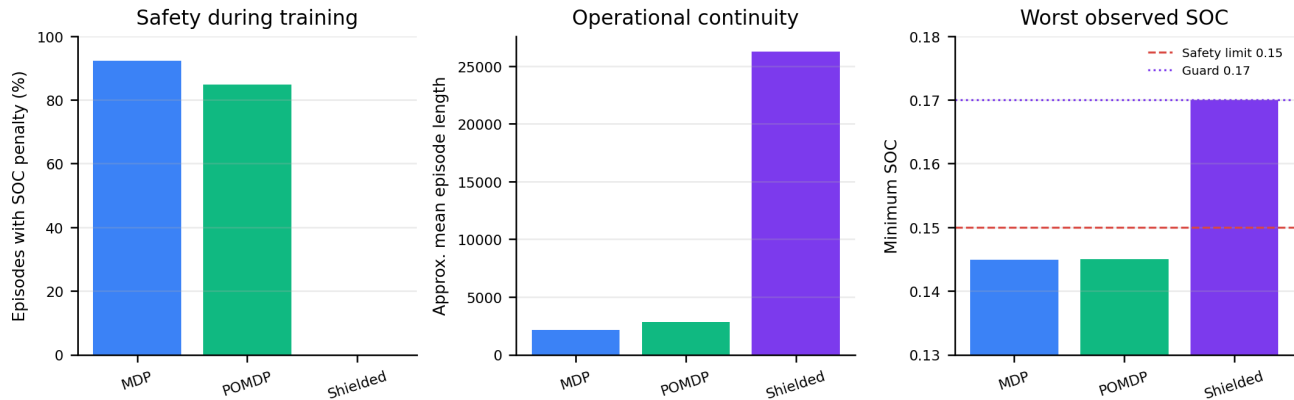


Figure 3. Aggregate training evidence from monitor and safety logs.

Metric	MDP-PPO	POMDP-LSTM-PPO	Shielded POMDP
Recorded episodes	231	177	19
Approx. mean episode length	2,165	2,825	26,279
SOC-penalty rate	92.21%	84.75%	No observed low-SOC termination
Minimum SOC	0.1449	0.1450	0.1700
Clipped/corrected events	878	290	16,628
Fully blocked actions	0	0	13,327
Evaluation behaviour	Peaked then collapsed	Positive checkpoint then degraded	Positive, mostly increasing

Convergence classification

MDP-PPO: non-convergent and safety-unstable.
 POMDP-LSTM-PPO: partially learned but operationally unstable.
 Shielded POMDP: operationally convergent with mild late-stage degradation.

Shielded training summary

19 full episodes; 499,301 monitored steps; 91.99% strict acceptance; 3.33% corrected; 2.67% blocked; zero fallback; minimum actual SOC = 0.17.

The flat 26,279-step episode-length curve is the strongest convergence evidence: the shield prevented catastrophic trajectories and supplied PPO with consistent full-horizon experience. The best evaluation checkpoint occurred near 450k steps rather than at the final checkpoint.

RELATIVE PROGRESSION FROM MDP TO POMDP

Indicator	Change after adding forecasts
SOC-penalty rate	92.21% → 84.75% (−7.46 percentage points)
Protection events	878 → 290 (≈67% reduction)
Approx. mean episode length	2,165 → 2,825 steps (≈30.5% increase)
Worst minimum SOC	0.1449 → 0.1450 (essentially unchanged)

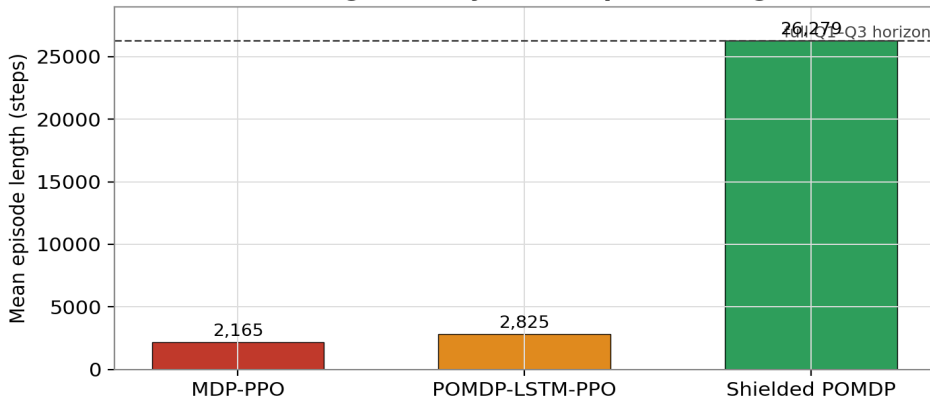
CHECKPOINT EVIDENCE

MDP evaluation peaked around 200k steps and later collapsed near −500. POMDP produced a useful intermediate checkpoint with reward 110.855, 1,999 steps and minimum SOC 0.153, but later degraded. Shielded evaluation remained positive, peaked near 450k around 138, and declined only modestly at 500k.

takeaway

LSTM forecasts reduced the frequency of risky behaviour, but only the shield changed the qualitative training pattern from repeated early termination to complete-horizon operation.

Training Stability: Mean Episode Length



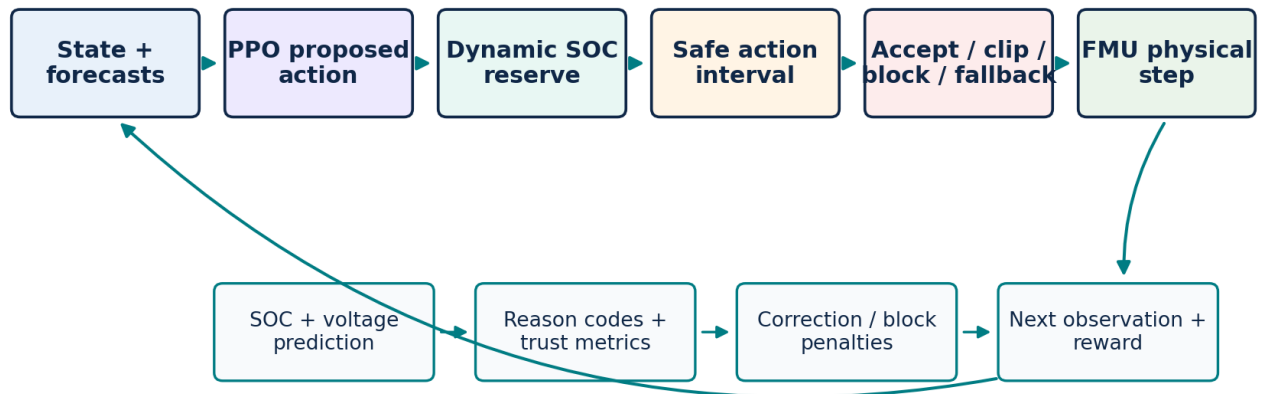
Controller	Classification
MDP-PPO	Non-convergent and safety-unstable
POMDP-LSTM-PPO	Partially learned but operationally unstable
Shielded POMDP	Operationally convergent (flat 26,279-step episodes)

07

Semi-Formal Runtime Safety Shield

EXPLICIT CONSTRAINTS, FORECAST-AWARE RESERVE AND INTERVENTION EVIDENCE

Policy-guided runtime safety shield



PPO optimizes performance; the shield enforces operational constraints before FMU execution.

Figure 4. The safety layer sits between PPO and the FMU and is evaluated at every 15-minute step.

SAFETY CONFIGURATION

Constraint	Value / rule
Hard SOC envelope	0.10–0.90
Reporting limits	0.15–0.85
Operational guard	0.17–0.83
Dynamic lower reserve	0.17 / 0.19 / 0.22
Voltage range	0.95–1.05 p.u.
Ramp limit	5 kW per step
Battery capacity	100 kWh
η_{charge} / $\eta_{\text{discharge}}$	0.95 / 0.95

INTERVENTION SEQUENCE

- Read current SOC, voltage, load, PV and forecasts.
- Raise the minimum reserve under high future load or low future solar.
- Intersect hardware, SOC, voltage and ramp-rate action bounds.
- Accept, clip, block or use conservative fallback.
- Re-check predicted next SOC and voltage before FMU execution.
- Log reason code, correction size, bounds and margins.

Why the guard can be “negative”

If the dynamic reserve suddenly rises to 0.22 while SOC is 0.17, the dynamic-reserve margin is -0.05 . This is a reserve deficit, not a violation of the 0.15 reporting limit. The

shield blocks further discharge and allows gradual recovery.

TERMINOLOGY AND SCOPE

The implementation is best described as a formalized-constraint or semi-formal runtime shield. It enforces explicit operational rules and prevented observed failures in the evaluated digital-twin scenarios, but it is not universally verified by reachability analysis, control barrier functions or theorem proving.

NUMERICAL SAFE-ACTION EXAMPLE

Given	Result
SOC = 0.18; dynamic reserve = 0.17; capacity = 100 kWh	Available energy above reserve = 1.0 kWh
$\eta_{dis} = 0.95$; $\Delta t = 0.25$ h	Maximum safe discharge ≈ 3.8 kW
PPO proposes +20 kW discharge	Shield clips to the safe upper bound
If reserve rises to 0.22 while SOC = 0.18	Discharge energy is zero; positive discharge is blocked

TRUST EVIDENCE LOGGED AT EVERY STEP

Proposed and executed actions, correction magnitude, safe bounds, predicted next SOC, predicted voltage, accepted/corrected/clipped/blocked status, forecast-transition warning, dynamic reserve, safety margins and intervention reason. These logs convert “trustworthy” from a qualitative label into measurable evidence.

takeaway

The runtime shield is a selective safety supervisor: it preserves PPO as the primary decision maker while independently enforcing SOC, voltage, ramp and forecast-aware reserve constraints.

08

Three-Day Behavioural Comparison

SAME ENVIRONMENT, DIFFERENT CONTROL PHILOSOPHIES

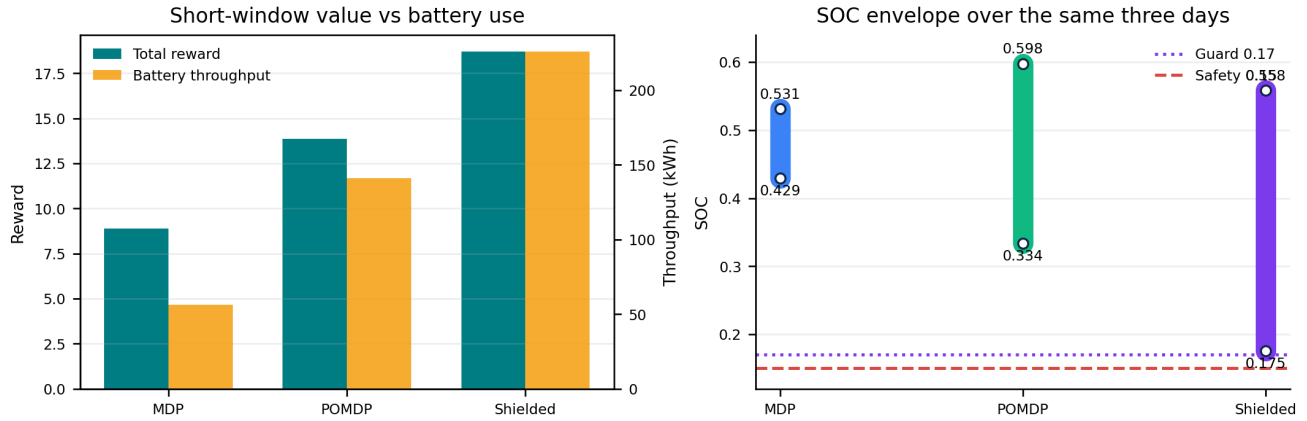
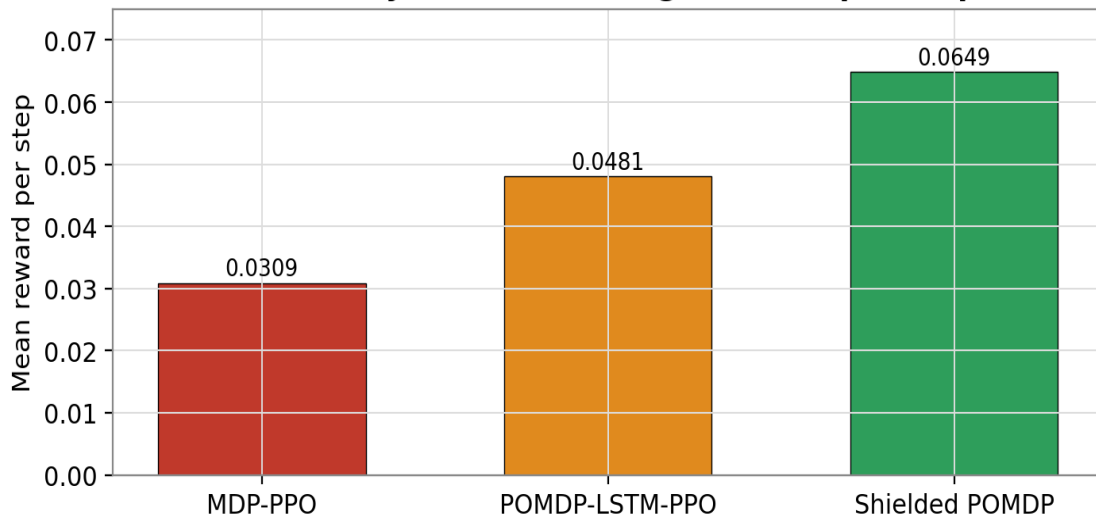
Three-day behavioural progression: conservative → forecast-aware → bounded high utilisation**Three-Day Window: Average Reward per Step**

Figure 5. Short-window reward, throughput and SOC envelope for the best evaluation checkpoints.

Metric	MDP-PPO	POMDP-LSTM-PPO	Shielded POMDP
Completed steps	288	288	288
Total reward	8.8875	13.8503	18.6972
Mean reward / step	0.0309	0.0481	0.0649
SOC range	0.4295–0.5314	0.3336–0.5977	0.1753–0.5582
Charge energy	30.42 kWh	75.55 kWh	119.47 kWh
Discharge energy	26.00 kWh	65.40 kWh	106.35 kWh
Throughput	56.42 kWh	140.95 kWh	225.82 kWh
Corrections / fallback	0 / 0	0 / 0	0 / 0

Controller behaviour

MDP uses the battery conservatively. POMDP cycles more strongly using forecast information. The shielded policy uses the widest safe SOC range and highest battery throughput.

Interpretation boundary

Zero correction in this window does not mean the shield was inactive: the dynamic reserve was active on 80.21% of steps and forecast-transition warnings occurred on 4.86%. PPO proposals simply remained inside the safe set.

RELATIVE SHORT-WINDOW OUTCOMES

Comparison	Reward change	Throughput change
POMDP vs MDP	+55.8%	≈+150%

Comparison	Reward change	Throughput change
Shielded vs MDP	+110.4%	≈+300%
Shielded vs POMDP	+35.0%	≈+60.2%

GRID AND BATTERY INTERPRETATION

MDP leaves more daytime PV exported and imports more at night because it uses a narrow SOC band. POMDP absorbs more surplus and schedules stronger daily cycles. The shielded controller narrows grid exchange further by exploiting the permitted safe envelope, but the additional throughput foreshadows the larger Q4 and annual heat-loss values.

SHORT-WINDOW LIMITATION

Three benign days cannot represent prolonged winter depletion, seasonal price variation, upper-SOC saturation, forecast error accumulation or annual degradation. The window is used for behavioural illustration only; complete-period rankings come from Q4 and annual tests.

takeaway

The short window visually demonstrates the methodological progression, but Q4 and annual evaluations remain the basis for safety, economics and deployability conclusions.

09

Q4 Unseen Winter Evaluation

COMPLETE-PERIOD RANKING AND LOW-SOC FAILURE DIAGNOSIS

Q4 unseen winter test: completion changes the interpretation of performance totals

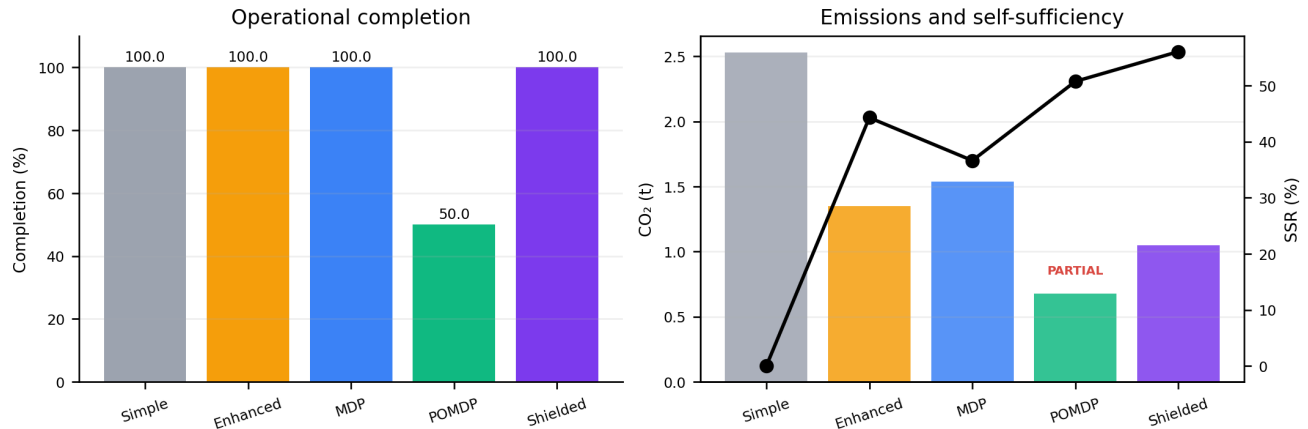


Figure 6. The unshielded POMDP values are partial-run totals because it completed only 49.95% of Q4.

Controller	Completion	CO ₂ t	SSR %	SSC %	Import kWh	Heat kWh	Net result
Simple RBC	100%	2.5305	0.00	6.65	7,282	394.26	\$458.34 cost
Enhanced RBC	100%	1.3481	44.28	66.28	4,037	117.01	\$168.05 cost
MDP-PPO	100%	1.5370	36.63	53.46	4,591	57.08	\$242.32 cost
POMDP-LSTM-PPO	49.95%*	0.6752*	50.74*	67.26*	1,968*	99.39*	\$5,085.76 cost
Shielded POMDP	100%	1.0476	56.05	84.74	3,184	237.71	\$92.35 cost

POMDP failure at step 4,374

SOC 0.14922 while still discharging +0.3243 kW; voltage 0.99838 p.u.; PV 0.717 kW; load 3.638 kW; stability_flag = 0. The evidence strongly associates the stop with lower-bound SOC operation.

Complete-controller result

The shielded controller delivered the lowest CO₂, voltage deviation, grid import and net cost, together with the highest SSR and SSC. Its cost reduction versus Simple RBC was 79.85%.

Trade-off: shielded heat loss was 237.71 kWh—lower than Simple RBC but about twice Enhanced RBC and more than four times MDP-PPO. Better local energy utilization was achieved through higher battery activity.

Q4 IMPROVEMENT AMONG COMPLETE CONTROLLERS

Improvement (complete controllers)	vs Simple RBC	vs Enhanced RBC	vs MDP-PPO
CO ₂ reduction	58.6%	22.3%	31.8%
Grid-import reduction	56.3%	21.1%	30.6%
Net-cost reduction	79.9%	45.0%	61.9%

Its low CO₂ and import values cover only the first 49.95% of Q4. Ratios such as SSR and SSC describe behaviour during the completed segment, but they do not establish winter reliability. The failure penalty correctly makes the incomplete controller economically unattractive.

Q4 RANKING

Composite result: 1) Shielded POMDP, 2) Enhanced RBC, 3) MDP-PPO, 4) Simple RBC, 5) unshielded POMDP after failure. Battery-health ranking differs: MDP-PPO and Enhanced RBC have lower heat loss.

takeaway

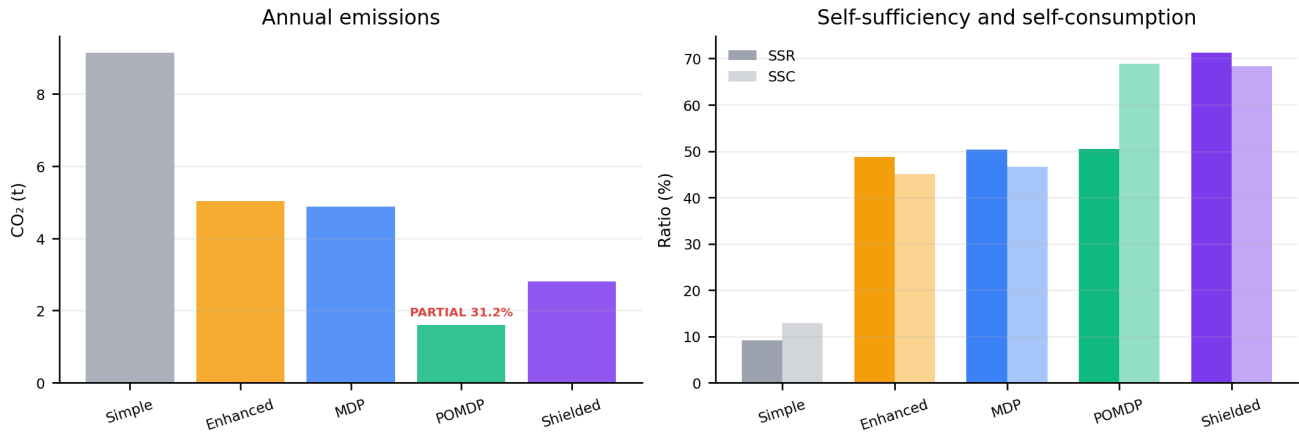
Q4 demonstrates the core safety case: forecast-aware performance is not deployable when one boundary violation can terminate the physical digital twin; the shield restores complete winter operation.

10

Full-Year 2023 Evaluation

ANNUAL PERFORMANCE, UPPER-SOC FAILURE AND BATTERY TRADE-OFF

Full-year 2023: compare complete controllers; treat POMDP values as partial-run diagnostics



Annual Energy Autonomy (Complete Controllers)

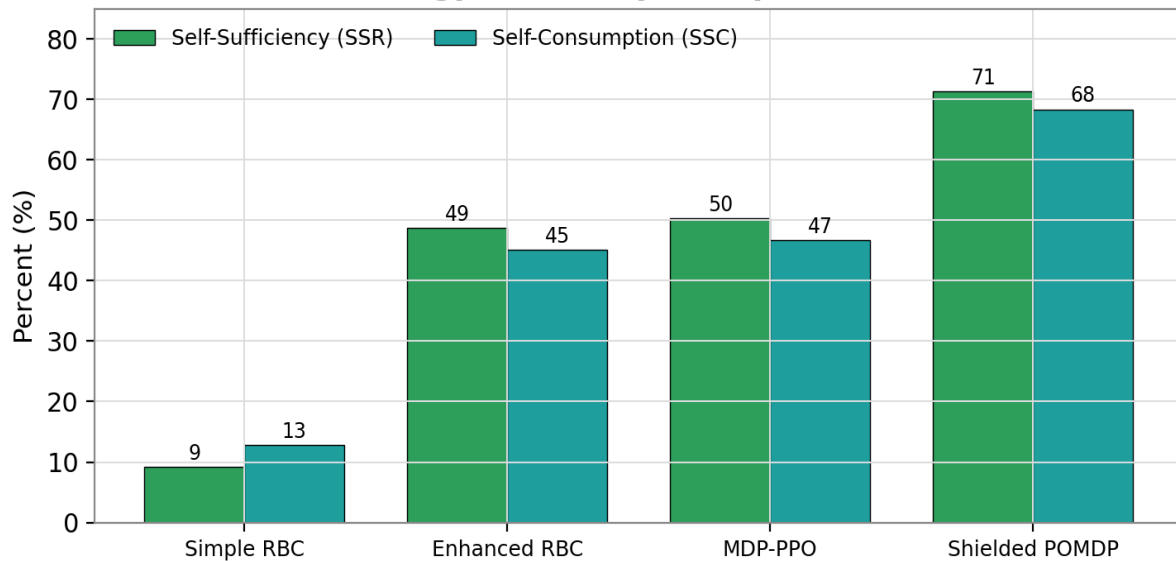


Figure 7. Annual controller comparison with the incomplete POMDP explicitly marked as a partial run.

Controller	Completion	CO ₂ t	SSR %	SSC %	Import kWh	Heat kWh	Economic result
Simple RBC	100%	9.1491	9.21	12.84	28,667	1,576.17	\$1,261.27 cost
Enhanced RBC	100%	5.0355	48.68	45.10	16,203	284.73	\$317.21 cost
MDP-PPO	100%	4.8878	50.33	46.69	15,684	350.59	\$348.05 cost
POMDP-LSTM-PPO	31.15%*	1.6055*	50.45*	68.84*	5,466*	239.16*	\$5,386.86 cost
Shielded POMDP	100%	2.8061	71.28	68.34	9,068	1,269.21	\$401.46 benefit

POMDP failure at step 10,915

SOC 0.85026 while still charging -0.1859 kW; voltage 1.00096 p.u.; grid export 1.530 kW; PV 6.654 kW; stability_flag = 0. This is an upper-bound failure, different from Q4.

Annual shielded result

Among complete controllers: 69.3% less CO₂ and 68.4% less grid import than Simple RBC; highest SSR and SSC; lowest voltage deviation; only controller with a modeled annual net benefit.

Battery trade-off: annual heat loss was 1,269.21 kWh—4.46× Enhanced RBC and 3.62× MDP-PPO, although still 19.5% below Simple RBC. The shielded policy occupies a different Pareto point rather than dominating every metric.

ANNUAL IMPROVEMENT AMONG COMPLETE CONTROLLERS

Baseline	CO ₂ reduction	Grid-import reduction	Voltage-deviation reduction
vs Simple RBC	69.3%	68.4%	≈66.7%

Baseline	CO ₂ reduction	Grid-import reduction	Voltage-deviation reduction
vs Enhanced RBC	44.3%	44.0%	≈44.0%
vs MDP-PPO	42.6%	42.2%	≈41.7%

COMPLETE-CONTROLLER RANKING BY OBJECTIVE

Objective	Strongest annual controller
CO ₂ , SSR, SSC, import, voltage and composite value	Shielded POMDP
Battery heat / stress proxy	Enhanced RBC
Adaptive low-throughput learning baseline	MDP-PPO
Conventional simplicity and interpretability	Enhanced RBC

Self-consumed PV reached 23,754.56 kWh for the shielded controller—about 51.5% above Enhanced RBC and 46.4% above MDP-PPO. This improved local utilization explains both the autonomy gains and the higher cycling burden.

The battery trade-off

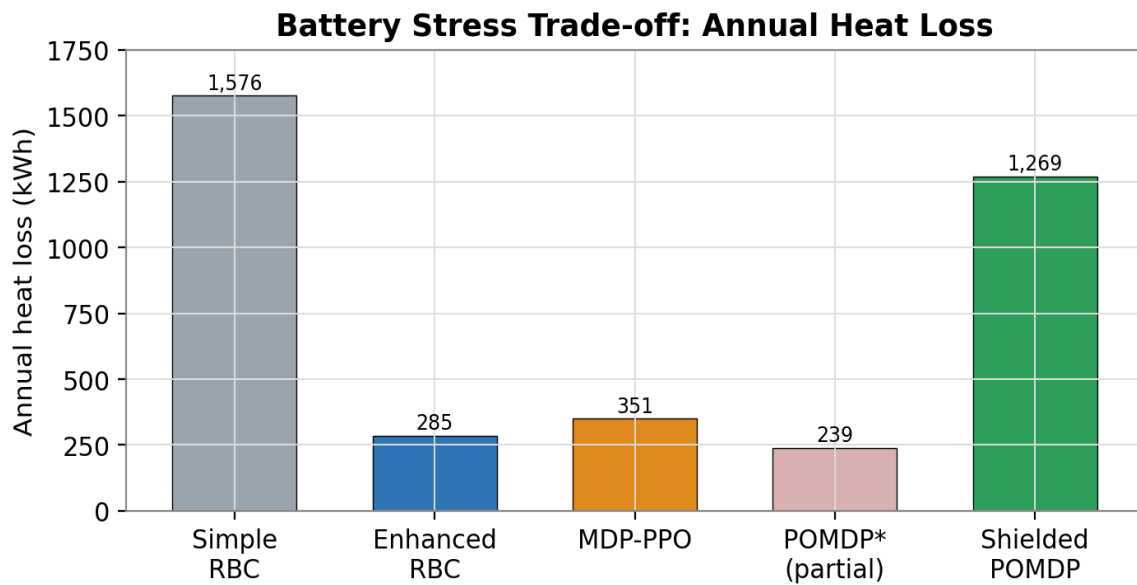


Figure 6. Annual heat loss. The shielded controller trades higher battery stress for better energy and economic outcomes.

Annual heat loss was 1,269 kWh — about 4.5× Enhanced RBC and 3.6× MDP-PPO, though still 19.5% below Simple RBC. Self-consumed PV reached 23,755 kWh, roughly 51% above Enhanced RBC. So the shielded controller occupies a different point on the trade-off curve rather than beating every metric.

takeaway

The annual evidence shows two unshielded POMDP boundary modes—continued discharge near the lower limit and continued charge near the upper limit—while the shield maintains full-year continuity. This is a Pareto trade-off, not domination. The shielded controller wins on emissions, autonomy, import, voltage and economics; the rule-based baselines win on battery stress.

Composite Techno-Economic Evaluation

COSTS, CREDITS, FAILURE PENALTY AND LOCAL RESILIENCE VALUE

Composite techno-economic value: failure penalties and local resilience value are explicit

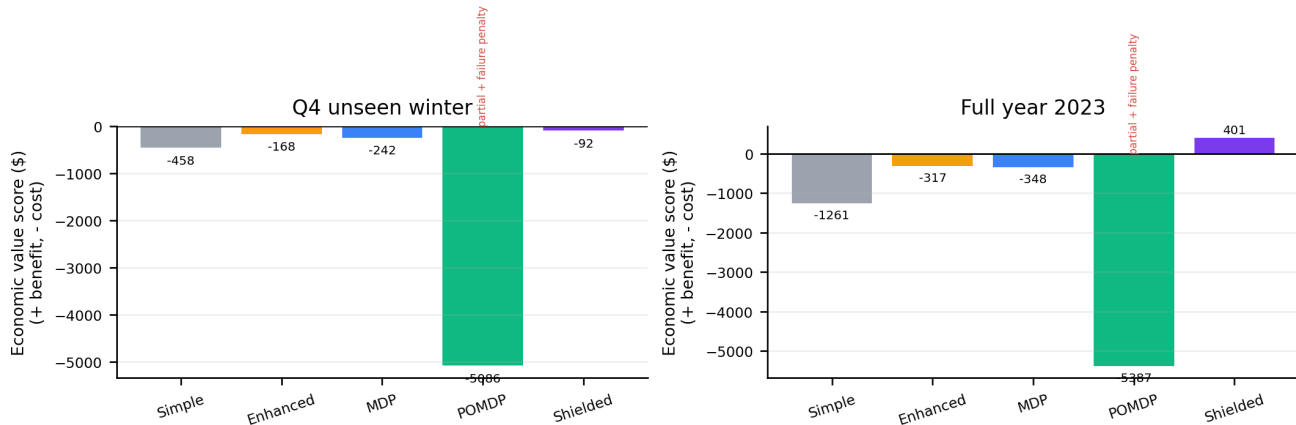


Figure 8. Economic value score: positive = modeled benefit; negative = modeled cost.

ANNUAL ECONOMIC COMPONENTS (\$)

Controller	Gross cost \$	Credits / value \$	Net result \$
Simple RBC	2,934.78	1,673.51	1,261.27 cost
Enhanced RBC	1,694.14	1,376.93	317.21 cost
MDP-PPO	1,799.55	1,451.50	348.05 cost
POMDP-STM*	5,779.14	392.28	5,386.86 cost
Shielded POMDP	1,170.04	1,571.50	401.46 benefit

MODEL CONTENTS

- Grid-import energy cost
- Grid-export market credit
- Local PV/resilience value
- Carbon cost
- Battery-wear cost
- Voltage-deviation cost
- Operational failure penalty

Sensitivity warning

About 75.6% of the shielded controller's total annual credit comes from the local resilience component. Without that assumed value, the result would be a net cost rather than a surplus.

INTERPRETATION AND RANKING

Period	Best complete result	Improvement vs Simple RBC	Important qualification
Q4	Shielded: \$92.35 cost	79.85% lower net cost	POMDP partial run receives failure penalty
Full year	Shielded: \$401.46 benefit	Removes \$1,261.27 cost + adds \$401.46 benefit	Composite value, not electricity bill only

For robust reporting, present sensitivity tests for the failure penalty and local resilience coefficient. Economic ranking should be based on signed net operating cost, not the absolute magnitude of the result.

ECONOMIC CALCULATION

Equation	Meaning
Gross cost = import + carbon + wear + voltage + failure	All positive operating burdens
Credits / values = export credit + local resilience value	Monetized system benefits
Net operating cost = gross cost – credits	Positive = cost; negative = benefit
Economic value score = – net operating cost	Higher is better for visualization

RECOMMENDED SENSITIVITY CHECKS

Assumption	Suggested range	Purpose
Failure penalty	\$1k / \$2.5k / \$5k / \$10k	Tests whether failed POMDP remains last
Local resilience coefficient	0% to current value	Tests whether annual surplus persists
Battery wear cost	Low / nominal / high	Tests the high-throughput shielded strategy

The reported annual shielded value is a composite techno-economic result, not a utility bill. The correct narrative is that modeled credits and local-energy value exceed modeled operating costs under the stated assumptions.

takeaway

The economic result rewards deployability: a controller that crashes is not allowed to appear attractive merely because it consumed less energy before failure.

12

Trustworthiness and Safety Evaluation

COMPLETION, VIOLATIONS, MARGINS AND INTERVENTION BURDEN

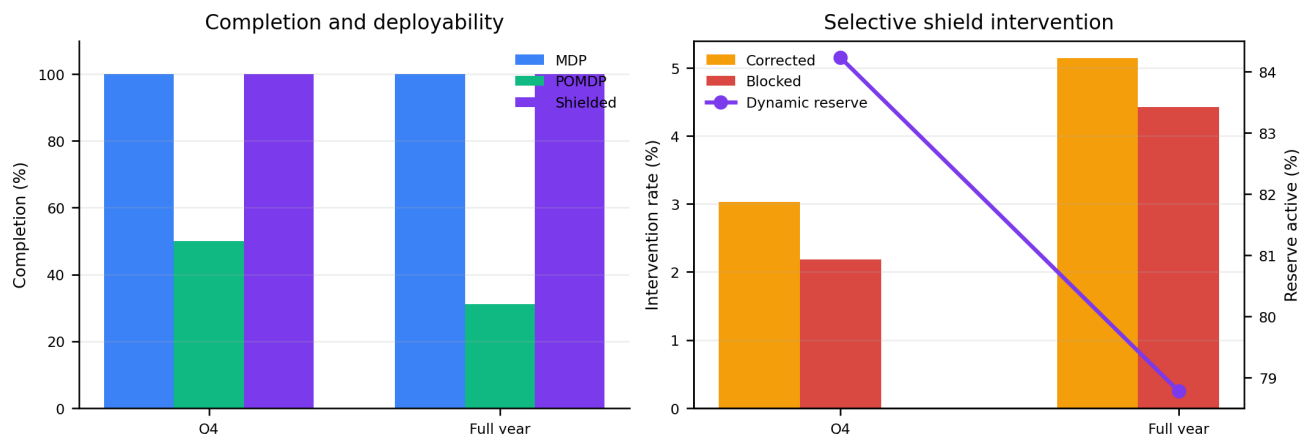
Trustworthiness is measured by completion, violations, margins and intervention burden

Figure 9. Completion and selective shield intervention across Q4 and the full year.

Period / controller	Completion	Violations	Stability failures	Min SOC	Corrected	Blocked	Dynamic reserve
Q4 MDP	100%	0	0	0.3469	N/A	N/A	N/A
Q4 POMDP	49.94%	1	1	0.1492	N/A	N/A	N/A
Q4 Shielded	100%	0	0	0.1700	3.03%	191 / 2.18%	84.23%
Annual MDP	100%	0	0	0.3416	N/A	N/A	N/A
Annual POMDP	31.15%	1	1	0.1765*	N/A	N/A	N/A
Annual Shielded	100%	0	0	0.1700	5.14%	1,551 / 4.43%	78.78%

Why one violation is severe

The POMDP violation rate is numerically tiny because only one 15-minute step was unsafe. Yet that single boundary crossing produced an FMU stability failure and terminated half of Q4 or two-thirds of the year.

Why MDP is an empirical pass

The selected MDP checkpoint completed both periods with comfortable SOC and voltage margins, but it has no independent runtime enforcement. Its safe behaviour is checkpoint- and scenario-dependent.

SHIELD INTERPRETATION

Accepted-action rate: 95.19% in Q4 and 91.36% annually. Fallback activations: zero. The shield therefore acted selectively rather than replacing PPO. Blocked actions are a subset of corrected/clipped actions and must not be added as independent categories.

METRIC DEFINITIONS AND FAILURE DEPTH

Metric	Interpretation
Unsafe duration	Number of violating steps $\times$ 0.25 h
Lower-depth %	$\max(0, 0.15 - \text{minimum SOC}) \times 100$
Upper-depth %	$\max(0, \text{maximum SOC} - 0.85) \times 100$
Voltage margin	Distance to the nearest 0.95/1.05 p.u. boundary
Correction rate	Fraction of shielded proposals numerically changed
Blocked rate	Fraction reduced to approximately zero

CRITICAL INTERPRETATION

Q4 POMDP violated the lower boundary by only about 0.078 percentage points, yet the simulation stopped. The annual failure occurred above the upper boundary, so minimum SOC alone would hide the failure. Both minimum and maximum SOC must be reported together with completion and stability failure count.

N/A FOR UNSHIELDED CONTROLLERS

Accepted, corrected, blocked, fallback and dynamic-reserve rates should be shown as N/A—not zero—for controllers without the formal shield. Zero would incorrectly suggest that a trusted layer checked and accepted every action.

takeaway

The final controller is evaluated not only by reward or cost, but by completion, violation severity, safety margins, intervention rate, fallback behaviour and stability of the physical digital twin.

13

Computational Overhead and Real-Time Feasibility

MEASURED ENVIRONMENT-TRANSITION LATENCY

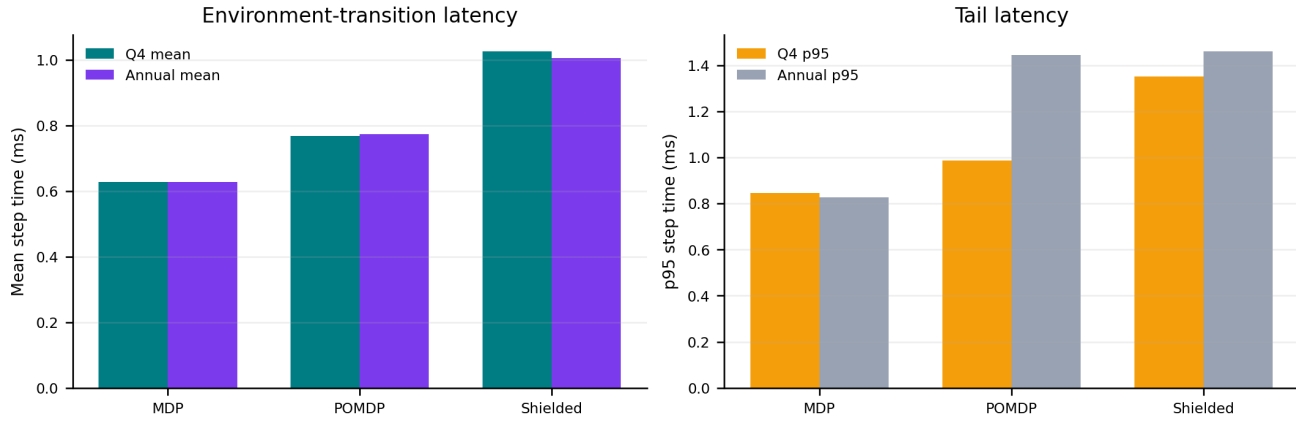
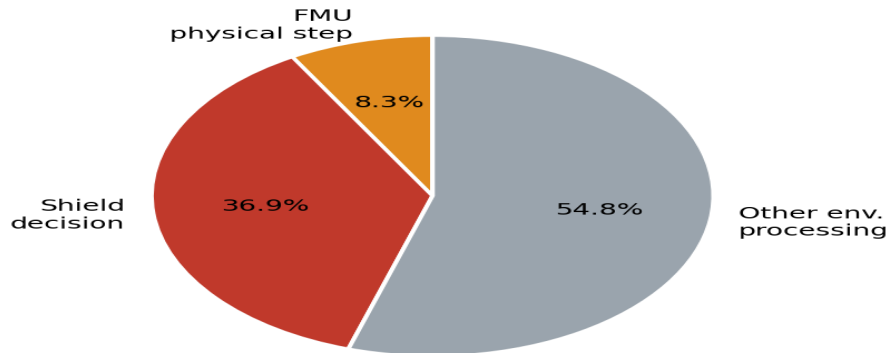
The shield adds ~0.37 ms of decision time within a 900,000 ms control interval**Shielded Step Time Breakdown (total ≈ 1.01 ms per 15-min step)**

Figure 10. Mean and p95 latency remain near 1 ms for the shielded environment.

Period	Controller	Mean total ms	FMU ms	Shield ms	p95 total ms	Approx. steps/s
Q4	MDP	0.6277	0.0750	—	0.8463	1,593
Q4	POMDP	0.7672	0.0855	—	0.9875	1,303
Q4	Shielded	1.0258	0.0815	0.3797	1.3512	975
Annual	MDP	0.6271	0.0751	—	0.8274	1,595
Annual	POMDP	0.7739	0.0905	—	1.4450	1,292
Annual	Shielded	1.0059	0.0836	0.3707	1.4598	994

Absolute overhead

The shield adds approximately 0.37–0.38 ms of decision time. Relative overhead is about 30–34% versus unshielded POMDP, but the absolute increase is only around 0.23–0.26 ms in total environment-step time.

Deadline margin

A 15-minute supervisory interval equals 900,000 ms. Mean shielded step time of about 1.01 ms uses only ≈0.000112% of the available interval; p95 remains below 1.46 ms.

MEASUREMENT SCOPE

The timing covers env.step(action): shield logic, FMU step, state extraction, reward and logging. It may exclude PPO inference and online LSTM inference; the current study uses precomputed forecast features. Communication, sensors, actuators and embedded hardware are also outside the measurement boundary.

FULL-YEAR TIMING DECOMPOSITION FOR THE SHIELDED CONTROLLER

Component	Mean time	Share / interpretation
FMU physical step	0.0836 ms	≈8.31% of total

Component	Mean time	Share / interpretation
Shield decision	0.3707 ms	≈36.85% of total
Remaining environment processing	≈0.5516 ms	State extraction, reward and logging
Total environment transition	1.0059 ms	≈994 simulated steps/s

APPROXIMATE TRANSITION RUNTIME

Evaluation	MDP-PPO	Shielded POMDP
Q4	≈5.50 s	≈8.98 s
Full year	≈21.97 s	≈35.25 s

WHAT IS NOT YET INCLUDED

PPO neural-network inference, online LSTM inference, sensor acquisition, communication delay, actuator response and embedded-hardware timing. The current result therefore supports supervisory real-time feasibility in simulation, not hard real-time certification on deployment hardware.

Professor-ready takeaway

The runtime shield restored complete operation while adding less than 0.4 ms of mean decision overhead per 15-minute control step in the evaluated digital-twin environment.

14

Explainable AI Layer and Fidelity Validation

GLOBAL DEPENDENCE, LOCAL EXPLANATIONS AND REWARD TRADE-OFFS

Improved XAI is quantitatively more faithful across all controllers

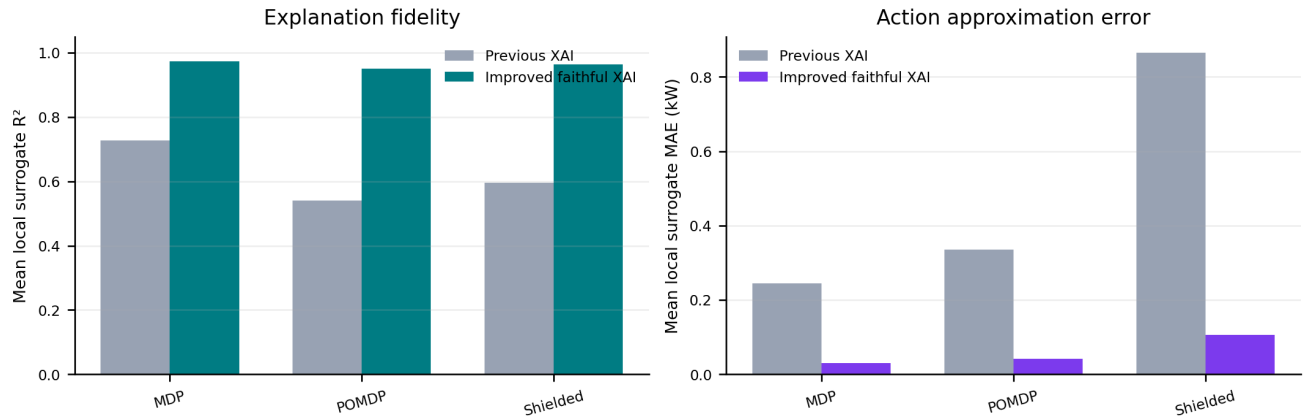


Figure 11. Improved faithful XAI substantially increases local surrogate fidelity and reduces action error.

Agent	R ² previous → improved	MAE previous → improved	Label agreement	Local stability
MDP-PPO	0.7280 → 0.9735	0.2454 → 0.0297 kW	95.93%	75.68%
POMDP-LSTM-PPO	0.5407 → 0.9498	0.3346 → 0.0414 kW	98.02%	90.60%
Shielded POMDP	0.5952 → 0.9635	0.8651 → 0.1063 kW	97.00%	90.93%

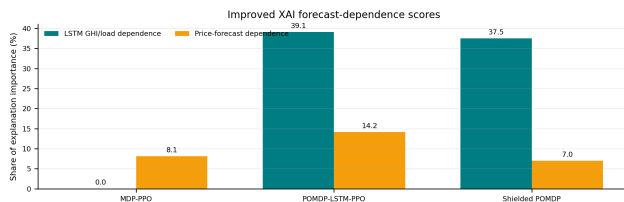


Figure 12. Forecast dependence estimated by the improved XAI method.

Local example at step 1,800

Action: charge -5.6507 kW. PV 11.809 kW, load 2.283 kW, grid export 3.875 kW, SOC 0.323. Fidelity R² = 0.9681; MAE = 0.0439 kW; label agreement and local stability = 100%.

HUMAN-READABLE INTERPRETATION

Large current PV surplus and near-term GHI forecasts pushed the policy toward charging. SOC, load and price created an opposing discharge incentive, but the renewable-surplus signal dominated. Local-energy and economic reward components were positive, while voltage and battery-health terms were small trade-offs.

Fidelity claim boundary

High local R² means the surrogate reproduces PPO behaviour inside the sampled neighbourhood. It does not mean PPO is globally linear or that the surrogate can replace the policy. Report neighbourhood design, held-out perturbations and repeated XAI seeds.

FORECAST-DEPENDENCE RESULT

Agent	LSTM dependence	Price-forecast dependence	Interpretation
MDP-PPO	0.00%	8.10%	Reactive to current physical state; no LSTM features
POMDP-LSTM-PPO	39.11%	14.16%	Most anticipatory policy
Shielded POMDP	37.51%	6.96%	Forecast-aware with greater current-state emphasis

LOCAL REWARD DECOMPOSITION AT THE ILLUSTRATED DECISION

Contribution	Value	Meaning
Local energy effect	+0.05708	Main benefit from absorbing PV surplus
Economic effect	+0.04597	Charging is economically favorable
Voltage effect	-0.00242	Small power-quality trade-off
Battery-health effect	-0.00071	Small cycling cost

VALIDATION CHECKLIST

Use held-out perturbations, identical explanation points, the same perturbation distribution, repeated XAI seeds and feature-rank stability. The shield explanation must also distinguish reserve activation from numerical intervention when proposed and executed actions are identical.

takeaway

The improved XAI layer does not assume explanations are trustworthy; it measures local fidelity, action error, directional agreement and policy stability alongside each explanation.

15

Integrated Comparison, Contributions and Limitations

FINAL PROFESSOR-READY SCORECARD

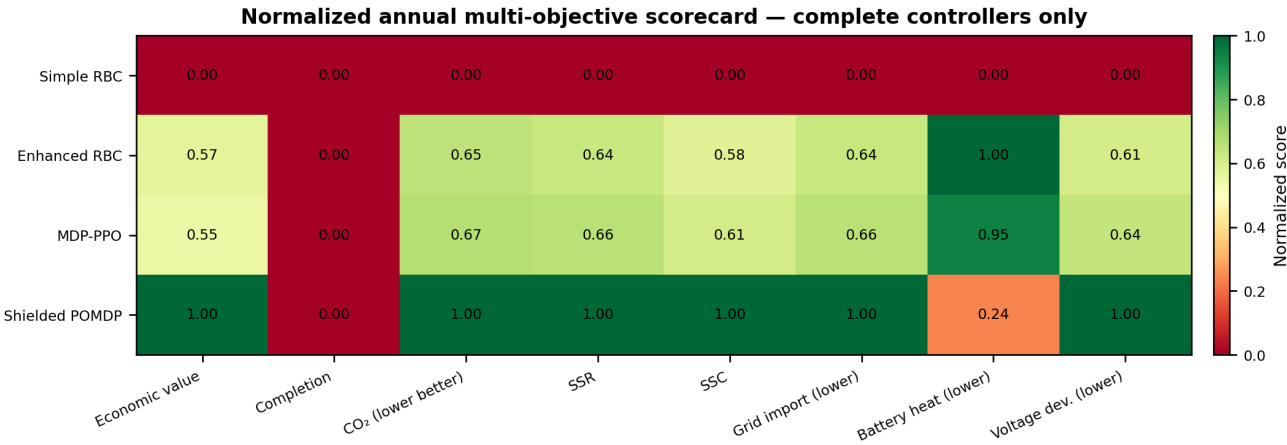


Figure 13. Normalized annual scorecard for complete controllers only; higher is better.

Controller	Completion	Annual CO ₂	SSR / SSC	Grid import	Heat loss	Economic result	Safety interpretation
Simple RBC	100%	9.149 t	9.21 / 12.84%	28,667 kWh	1,576 kWh	\$1,261 cost	Stable but inefficient
Enhanced RBC	100%	5.036 t	48.68 / 45.10%	16,203 kWh	285 kWh	\$317 cost	Strong conventional baseline
MDP-PPO	100%	4.888 t	50.33 / 46.69%	15,684 kWh	351 kWh	\$348 cost	Empirical safe completion
POMDP-LSTM-PPO	31.15%*	1.606 t*	50.45 / 68.84%*	5,466 kWh*	239 kWh*	\$5,387 cost	Boundary failure; partial run
Shielded POMDP	100%	2.806 t	71.28 / 68.34%	9,068 kWh	1,269 kWh	\$401 benefit	Full completion with enforcement

SCIENTIFIC CONTRIBUTIONS

- Unified Modelica FMU–Python–Gymnasium ECPS loop.
- Forecast-aware PPO observation and dynamic reserve.
- Selective runtime safety enforcement with reason codes.
- Crash diagnostics at low- and high-SOC boundaries.
- Integrated physical, economic, trust, overhead and XAI evaluation.

LIMITATIONS AND NEXT EVIDENCE

- Single PPO seed in the present headline results.
- One-year, one-climate evaluation.
- Composite economics depends on resilience and failure assumptions.
- Online LSTM inference and hardware communication not timed.
- Shield is semi-formal, not universally verified.
- Battery degradation represented mainly through heat/throughput proxies.

Final conclusion

Results suggest that LSTM forecasts improve anticipatory dispatch, while runtime constraint enforcement is the decisive component for operational reliability. Among complete controllers, the shielded POMDP delivered the strongest overall techno-economic and environmental outcome in the evaluated digital-twin scenarios, with a clear battery-stress trade-off.

EVIDENCE-TO-CLAIM MAP

Claim	Direct evidence	Careful wording
Forecasts add useful information	Q4 error improvement and XAI dependence	Forecast-aware, not perfectly predictive
PPO adapts dispatch	Reward, SOC and action behaviour	Useful learned behaviour, not guaranteed optimality
Shield improves reliability	Completion, zero violations, correction/block logs	More operationally reliable in evaluated scenarios
Framework is computationally feasible	≈1 ms environment transition	Supervisory feasibility, not hardware certification
Explanations are faithful locally	R ² , MAE, label agreement and stability	High local fidelity, not global policy equivalence

PRIORITY NEXT EVIDENCE

Multiple PPO seeds with mean ± standard deviation; threshold and resilience-value sensitivity; multi-year and multi-climate tests; online LSTM/PPO end-to-end latency; additional Modelica diagnostic flags; and stronger formal verification through barrier functions, reachability or invariant-set analysis.

Professor-ready takeaway

Prediction improves awareness; PPO provides adaptive optimization; the runtime shield converts the learned policy into a more operationally reliable controller by monitoring, correcting and explaining actions before physical execution.